

Class 6: Linear Transformations

Up to this point we have thought of the expression $A\mathbf{x}$ as a linear combination of the columns of A .

In this class, when A is an $m \times n$ matrix, we will think of $A\mathbf{x}$ as the result of a function that takes vectors in $\mathbf{R}^n$ and transforms them into a vector in $\mathbf{R}^m$.

Example $A = \begin{bmatrix} 1 & 3 & 2 & 8 \\ 2 & 2 & 0 & 4 \\ 3 & 1 & -2 & 0 \end{bmatrix} \quad T(\mathbf{x}) = A\mathbf{x}$

a. $T\left(\begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}\right) =$

b. $T\left(3\begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}\right) =$

c. Is there a vector $\mathbf{x}$ such that $T(\mathbf{x}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

In other words

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Key Idea: range of T = column space of matrix A

Is all of $\mathbf{R}^3$ in the range of T ?

Key Idea: Defining property of any linear transformation

A linear transformation T is any function that satisfies the following two properties.

1. T preserves addition
2. T preserves scalar multiplication

Consequences of this

- 1.
- 2.
3. If A is any matrix and we define $T(\mathbf{x}) = A\mathbf{x}$ then T will be a linear transformation.
4. Linear transformations always transform lines into lines.